Ridwan O. Olabiyi **Cover Letter**

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Dear Members of the Faculty Search Committee,

I am writing to apply for a tenure-track faculty position in Industrial and Systems Engineering. I am a Ph.D. Candidate in Industrial Engineering at Arizona State University, advised by Dr. Ashif Iquebal, and I expect to complete my Ph.D. in Spring 2027. My research develops uncertainty-aware scientific machine learning methods for inverse problems, governing physics discovery, and efficient computation in engineering systems. Across this work, I combine Bayesian inference, uncertainty quantification, explicit modeling of stochasticity, operator learning, causal reasoning, and physics-based simulation to develop models that remain reliable under sparse measurements, intrinsic variability, changing operating conditions, and the high cost of repeated high-fidelity computation.

My doctoral work provides the foundation for an independent research program centered on three connected areas. The first develops uncertainty-aware inverse methods for recovering hidden physical quantities and material properties from limited observations. This includes my Bayesian spatiotemporal inverse heat conduction work published in the *ASME Journal of Heat and Mass Transfer* and ongoing stochastic inverse modeling for elastoplastic material characterization with collaborators at NIST and Oak Ridge National Laboratory. The second focuses on discovering governing structure from uncertain data. My 2026 *Communications Physics* article, "Discovering Governing Physics in the Presence of Uncertainty," develops a stochastic framework for equation discovery under measurement noise and intrinsic variability. The third develops scientific machine learning for efficient engineering computation, including CRONet for accelerated topology optimization and related hardware acceleration work. I am also extending these ideas to image based 3D reconstruction for additive manufacturing quality assurance. As a faculty member, I plan to connect these areas through adaptive models that quantify uncertainty, recognize model inadequacy, and guide the next informative measurement, simulation, or experiment, with applications in manufacturing, materials, semiconductor, thermal, and energy systems.

I have also developed experience in research development and proposal preparation. I have contributed to grant writing with my advisor, including an NSF CAREER proposal and other federal funding efforts, which has given me practical experience in framing research significance, integrating research and education, developing work plans, and communicating broader impacts. My selection as a 2025 Purdue Trailblazers in Engineering Fellow further strengthened my preparation for faculty responsibilities. The program included intensive faculty development in strong application packages, academic career pathways, and early-career research funding through agencies such as NSF, AFOSR, and ONR. It also broadened my experience with grant development, course planning, class coordination, and the integration of research and teaching.

Teaching and mentoring are central to the faculty career I am pursuing. My teaching experience includes IEE 380 *Probability and Statistics for Engineering Problem Solving*, IEE 470 *Quality Control*, and IEE 570 *Advanced Quality Control* at Arizona State University, in addition to earlier mathematics, physics, science, and advanced-level courses instruction in Nigeria. My teaching philosophy begins with purpose and active engagement. I introduce technical ideas through the engineering questions and decisions they enable, make expectations and reasoning visible, and use authentic projects to help students progress from applying procedures to investigating problems independently. My mentoring spans undergraduate, master's, capstone, recent graduate, and doctoral researchers, including work in additive manufacturing quality assurance, active learning, generative CAD modeling, inverse characterization, and scientific machine learning.

My broader record reflects a commitment to collaborative research, professional service, and student development. I have served in leadership roles in the ASU INFORMS Student Chapter, organized and chaired conference sessions, reviewed for journals and conferences, and contributed to educational outreach. My recent recognitions include First Place and Best Presentation Design in the 2026 NSF Future Manufacturing Data Challenge, a winning solution in the 2026 Semiconductor Solutions Challenge, selection for the 2026 INFORMS Job Market Showcase, the 2025 Purdue

Trailblazers in Engineering program, and recognition as a 2024 INFORMS QSR Best Student Paper Competition finalist.

I would welcome the opportunity to contribute this combination of methodological research, collaborative engineering applications, engaged teaching, mentoring, and service to an Industrial and Systems Engineering program. Additional information on my research, publications, teaching, and professional activities is available on my academic website. Thank you for your time and consideration.

Sincerely,

Ridwan O. Olabiyi
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